

Hybrid Continuum-Tentacle Robotic Manipulator

Circuit Schematics and Preliminary Analysis

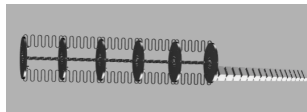
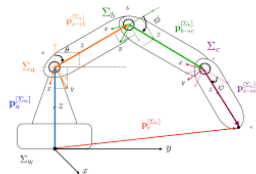
Veeresh S K, Rishith Gupta, Arshit Singh
Prithvi Sangam Chakravarthy, Arijit Paul

TUE-16
S-08

April 14, 2026

Problem Definition and Proposed Approach

- Traditional rigid manipulators struggle in **narrow and constrained environments**
- Applications such as search & rescue and industrial inspection require flexible robots
- A continuum manipulator has a continuously bending backbone rather than rigid joints
- We propose a **tendon-driven continuum** manipulator for smooth and adaptable bending
- A tentacle-like adaptive end-effector enables grasping objects of different shapes

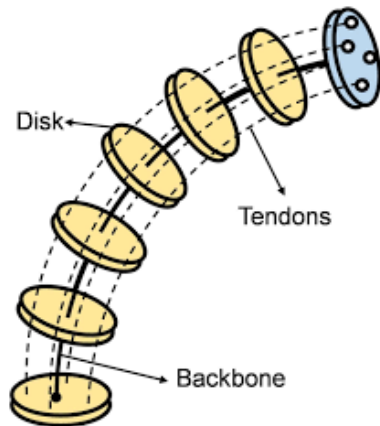


The robotic system consists of the following main subsystems:

- Continuum Link Mechanism
- Tentacle End-Effector
- Electronics : Components and Integration
- Algorithm and Controls System

Continuum Link

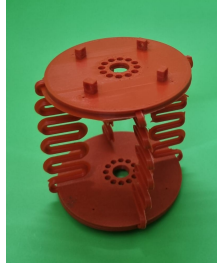
- Tendon-driven continuum backbone with a modular vertebrae structure
- Compliant structural design enables multi-directional bending motion
- Proposed backbone material: **hot glue stick** (high compressive strength with low bending stiffness for flexibility)
- Three design prototypes developed



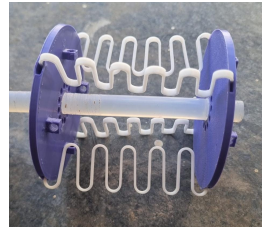
Continuum Link Prototypes



Prototype 1: Initial backbone design ;
Too sharp Edges, Low Compressive
strength



Prototype 2: Improved vertebrae
structure ; Low Compressive Strength



Prototype 3: No Problems as of now

Tentacle End-Effector

- Bio-inspired flexible gripper for object grasping
- Initial concept: single-body TPU print, but reliable printing proved difficult initially
- Adopted a **segmented hinge-based mechanism** to achieve controlled flexibility
- Tested multiple materials: Resin (too brittle) and PLA (friction issues)
- Implemented design adjustments to mitigate friction in PLA and improve motion



Tentacle End-Effector Prototypes



TPU Single-body Prototype
; Not printed properly



Hinge Mechanism Design
(PLA) ; Friction and uneven
tension causes to give
undesirable start position

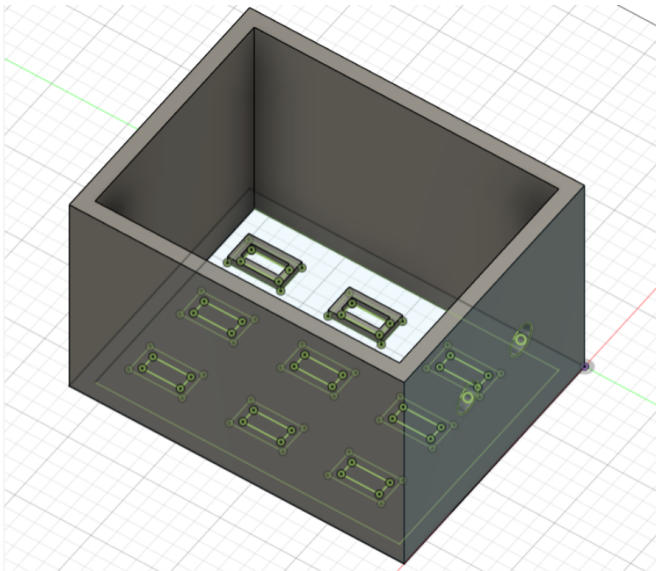


Resin Printed Prototype ;
Too brittle



PLA Prototype working with
adjustments

Base Plate

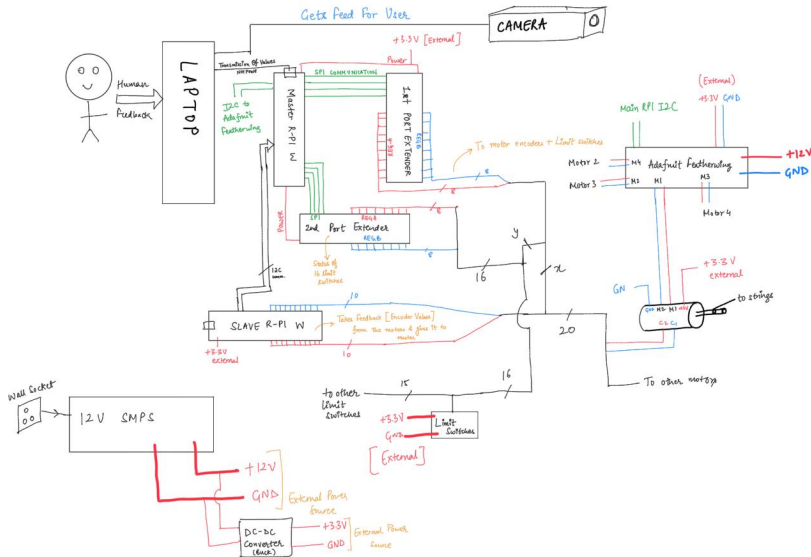


Electronics: Components and Integration

- Vision and User Input: Camera feed is streamed to the laptop, allowing the operator to issue motion commands based on live visual feedback
- Control Architecture: User inputs are sent to Raspberry Pi Pico W controllers, which process commands and drive the Adafruit FeatherWing motor driver
- Sensing and Feedback: Motor encoders and limit switches are read through Raspberry Pi GPIO pins and two port expanders for additional input channels
- Power System: A 12V SMPS powers the motors while a 3.3V regulator provides logic power for the control electronics

Next slide: Circuit diagram

Circuit Diagram

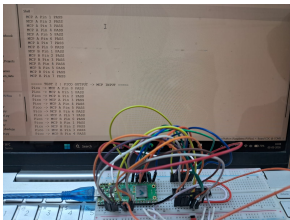


Justification for Sensors and System Components

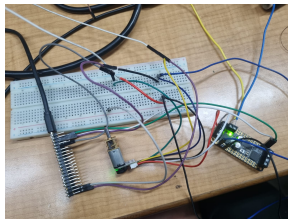
- **Limit Switches:** Detect contact and enforce motion limits for safe operation
- **Camera:** Provides real-time visual feedback for operator-guided teleoperation
- **Motor Encoders:** Provide position feedback for monitoring motor rotation
- **Motors:** Generate tendon tension to produce controlled robot bending
- **Adafruit FeatherWing:** Motor driver for controlling motor speed and direction
- **Raspberry Pi Pico W:** Main controller for command processing and actuation
- **12V SMPS & 3.3V Regulator:** Provide power for motors and logic electronics

Electronics Testing

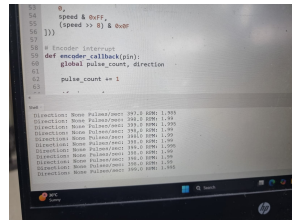
All Components individually tested



Port Expander testing



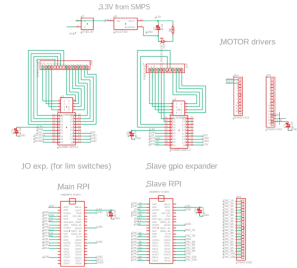
Testing RPi - Adafruit - motor - RPi
loop



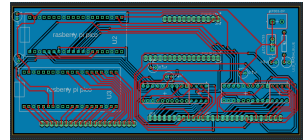
Getting Encoder values from motor

Schematics and PCB

- Circuit schematics were developed through multiple iterative design revisions
- The PCB design is currently undergoing iterative refinement to improve routing, integration, and overall reliability



Circuit schematic design



PCB layout iteration

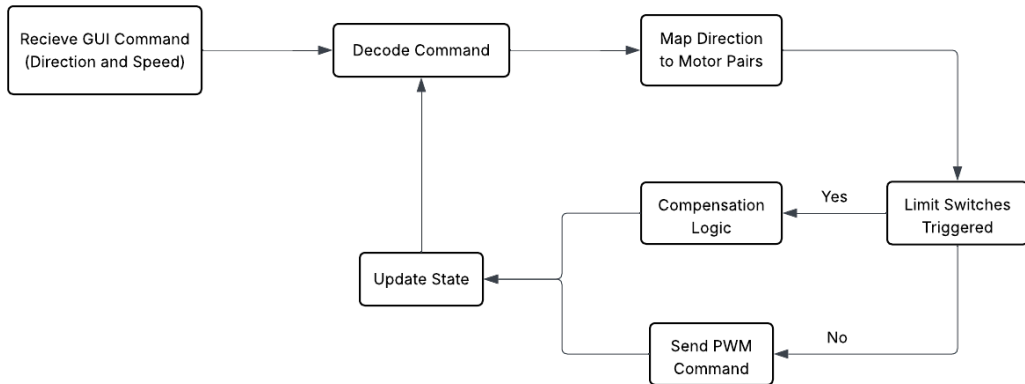
System Control Architecture

- GUI sends direction and speed commands to the Master RPi via UART
- The Master processes commands and determines motor actuation signals
- PWM signals drive motor controllers, actuating the tendon-driven continuum links
- Encoder feedback from the Slave RPi is sent to the Master RPi for monitoring

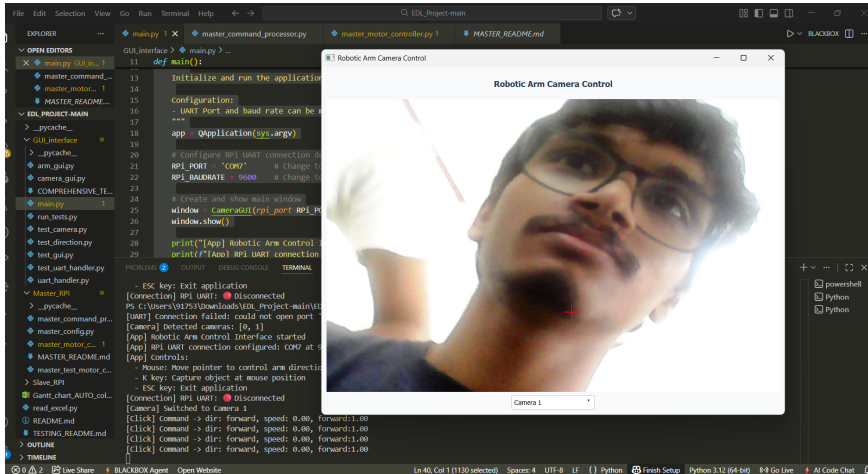
Key Control Features

- **Closed-loop control** with encoder feedback at 10 Hz for real-time monitoring
- Limit-switch safety constraints prevent motion beyond mechanical limits
- **Compensation logic** redistributes actuation when a link's motion is restricted
- Differential motor actuation enables coordinated bending of continuum links

Control Algorithm Overview



Algorithm and Controls System



Using the endoscope for camera feed and testing transmission to Microcontroller

Mechanical Development

- Finalize continuum link and tentacle end-effector CAD design and geometry
- Fabricate updated mechanical prototypes using WEL 3D printing facilities

Electronics Development

- Complete PCB design iterations and finalize the schematic and layout
- Fabricate the final PCB using the IIT Bombay PCB Lab facilities

Control Software Development

- Implement the control algorithm and motor actuation mapping in **MicroPython**
- Develop firmware for both Master and Slave Raspberry Pi control units

- **Hybrid Continuum–Tentacle Robotic Prototype**

A complete robot demonstrating continuum bending and adaptive object grasping

- **Navigation and Retrieval Demonstration**

Robot moves through a constrained path and retrieves a target object

- **User-Controlled Operation**

Robot movement controlled through GUI commands with live camera feedback

- **Integrated Mechatronic System**

Combined operation of mechanical structure, electronics, sensors, and control